

# INST 314 week 1: Data Basics

Dr. Scott Jackson

## What's in this document?

This is the primary reading for the first week of the course. It was written using R Markdown, so I am providing both the (easier-to-read) PDF rendering as well as the R Markdown “source”, which has all of the code examples and so on, if you find that to be useful.

The overall goal of this reading is to introduce some fundamental concepts that we will expand on throughout the course. In other words, the goal is to get us started off on some common ground.

I will call attention to key terminology with margin notes. I will also ramble about side thoughts in margin notes as well. Sometimes the margin notes will be numbered,<sup>1</sup> like footnotes that connect to a specific part of the main text. Sometimes they're unnumbered, just hanging out whenever you feel like reading them.

I will not always go into detail on the R code examples, because this is more of a narrative document, and I will explain details of all R code in other documents.

<sup>1</sup> This is a numbered side-note.

This is a margin note.

## Why statistics?

Before you start anything, it's always good to revisit the question of “why”, even if you think the answer is obvious (*spoiler*: there are almost always non-obvious things to think about).

This is a course about *statistics*, so why should we bother learning about statistics?

As with many good questions, when we consider them carefully, they lead to other questions. We might say something like, “statistics is an important part of data analysis,” and this is true, but it also raises another question, “what’s data analysis?”, and yet another, “what’s *data*?”

So let’s take that as our starting point: **What is data?**

There's another usage of the word "data" to represent *any* kind of digitally-recorded value.

The point for us here is to remember that we will use the word "data" in this course as defined

Take a few seconds to think about it, and write down or type out a list of things you think count as data. Yes, I’m serious, just do it. When you look at the list, what do you see? What seems to be in common? What isn’t? As I go through my definition below, think about whether you agree, and whether you think everything on your list counts as data under this definition. Should we make some alterations to the definition? These are all great things to discuss.

## Defining *data*

Excursion: is the word "data" plural or singular? Should we say, "data **is** important," or "data **are** important"? But there are also people who remind us that originally, the word "datum" meant "single value". I'm just warning you that "data are" snobs exist.

My definition has two parts. First, data represents a set of *values*. Those values can be numerical, alphabetical, or really any kind of symbolic value (a value represented by a symbol).

The crucial second part of the definition is that data must carry some kind of *meaning* or *information*. In turn, both of those words can have some pretty technical definitions, but we won't go down those rabbit holes for now. The point is that you could find a big list of numbers somewhere, but it's not really "data" until those numbers *mean* something, like if you found out that they were a set of financial records or game scores or medical measurements or something.

Put these together, and we can say something like, "data is a set of meaningful values," which is almost there, but not quite. Because think about it a little bit: if someone gave you a big file full of medical data, would we automatically say, "oh, that's a lot of valuable meaning in there"? If it's just a big collection of pulse readings without any connection to patients or other data, then probably not.

## Data analysis

Here's where we can bring in the idea of *data analysis*. Data may have some meaning, but I would argue that that meaning is not *accessible* without data analysis. We might also tweak our definition a bit to say something like "data is a set of values that *carry* meaning", with the corollary that data analysis is necessary in order to *extract* meaning from data.

[Clive Humby](#) famously said in 2006, "data is the new oil!" This metaphor has a lot of interesting facets if you think about it,

and I think the analogy between data as *unrefined* oil is apt here. Crude oil is of course necessary for most of the cars we drive, but only after a great deal of processing. The same with data: there may or may not be valuable information in the data, but only good analysis can reveal it or make it accessible or useable.

In this course, we will learn and practice both *exploratory* and *inferential* analysis.<sup>2</sup> These types of analysis are best done together, or are best considered as two useful but complementary approaches. We will discuss more about these as we go through the course, but as an initial working definition, *exploratory* analysis is a good first step, where we start to build an understanding of a data set, and start to develop questions about it. In contrast, *inferential* analysis is what we do when we would like to be a little more certain in what we can and can't conclude based on a data set.

<sup>2</sup> It is also common for people to contrast “exploratory” with “confirmatory.” I’m going to avoid that term until much later, because it’s more narrow.

## Embracing uncertainty

We said the “c-word” (“certain”), so now it’s time to discuss the opposite: *uncertainty*. Because this is what the core of statistics is really about. Let’s imagine a situation (obviously not that hard to imagine) in which there is a terrible new disease, and we would very much like to have a treatment that works to improve people’s outcomes. How do we go about doing this?

Well, we try things and we collect data. We take measurements about what we try, who we try it on, and what happens after people try our treatments. And let’s say that when I try a treatment on 10 people, 8 of them recover completely. How *certain* can I be that my treatment helped? The numbers sound encouraging, but what if I knew that out of 1,000 people who didn’t get any treatment, 780 of them recovered completely? If I just extrapolated my numbers, I might expect 800 people to recover out of 1,000. Can I say that my treatment would have saved 20 people out of 1,000? Can I say that with any certainty?

It turns out that in the real world, there are many random and unknown factors present in virtually everything we observe.

And sometimes, “dumb luck” just happens. If I flip a coin 10 times, on average I expect 5 heads and 5 tails, but what if I got 8 heads? Is that a sign that there’s something wrong with the coin, or was it just lucky/unlucky that I got 8 heads out of 10?

These are not esoteric questions of strange and unusual happenings. In the real world, we are perpetually surrounded by uncertainty. This can of course be worrisome and we may wish we had more certainty. But *pretending* to have certainty where you do not is much more problematic. Think about how many people seem absolutely certain about flawed ideas or misinformation they may see online. Confidence is a nice personal quality, but *overconfidence* in decision-making is just a way to make worse decisions.

However, the situation is not hopeless. People frequently mistake the mere presence of *any* uncertainty with the idea that there’s *zero* certainty. This not true. Just because you are only 90% sure of an outcome doesn’t mean it doesn’t matter!

So what can we do? My personal belief is that the best we can do is to **embrace uncertainty**. What does that mean? It means that we start by accepting that uncertainty is the rule, not the exception, and that our goal is to *understand* uncertainty, and do our best to *quantify* it, so that we can make the best decisions we can. Because when you accept that uncertainty exists, it changes your outlook to look for things that will help you *decrease* your uncertainty, without being under the illusion that you will ever get to *zero* uncertainty.

## Statistics

The good news is that there is an entire field devoted to better understanding and quantifying uncertainty, and this is what we know as *statistics*. When we talk about the “inferential analysis” above, we mean using statistics to quantify how much uncertainty we have and to help us identify what our “best bets” are.<sup>3</sup>

Let’s go back to the medical data example. Maybe we have a data set with a bunch of values of different kinds, linked in

<sup>3</sup> But also, once again, words and terms can end up with different meanings. In some usages, the word “statistic” can really just mean any kind of computed value. For example, the minimum, maximum, mean, median, etc. of a set of values are often called “descriptive statistics.” But this is not the same as what we mean by “statistical analysis,” which is more about how to draw conclusions. As a field, statistics has a *lot* of unfortunately confusing terms. It’s unfortunate because we have to learn to live with them, at least when we talk to statisticians.

different ways. This meets our definition of data, because each value carries some kind of meaning, because we know if it's a particular measurement, or whether it comes from a particular patient, and so on. But this data doesn't *tell* us anything.<sup>4</sup> But maybe we start to explore this data and we realize that there is a subgroup of people in the data who seems to be at higher-than-expected risk for a particular disease.

But this is still exploratory. If you look long enough in any data set, you may be able to find *some* kind of pattern, because humans are inherently good at perceiving (or thinking they perceive) patterns. At the end of the day, we would like to be able to properly *infer* something from the pattern, so that we can make better decisions. Back to our example, maybe we find out that there's an association between a group of people and a disease, but what we really want is to know, "okay, if you're a person in this group, does that raise your risk of this disease, independent of other factors?" In other words, can we use this information to make accurate predictions? How certain can we be that being in this group raises your risk?

This is the role of statistics. Valid statistical methods are mathematical methods that can help support drawing valid conclusions or making valid decisions by quantifying uncertainty. Notice I said *valid* statistical methods. Sloppy stats lead to sloppy or invalid conclusions. The author Tyler Vigen maintains this hilarious site of "spurious correlations."

<https://tylervigen.com/spurious-correlations>

These are all examples of things that, if you just run statistics on them, show very high correlations with each other. This is a reminder that statistics are not a substitute for thinking! They are best used as a *complement* to good thinking.

So it may be true that the frequency of Google searches for 'how to hide a body' significantly correlates with the number of Bachelor's degrees awarded in Library Science,<sup>5</sup> and the statistics help quantify how close that relationship is, but serious thought is needed to help make sure we are not blindly trusting stats, at least without a lot more careful work.

<sup>4</sup> This is why I hate the phrase "let the data speak for itself." Data doesn't say *anything* without an analysis, and people have to make choices when it comes to deciding exactly what to analyze and how.

<sup>5</sup> [https://tylervigen.com/spurious/correlation/1840\\_bachelors-degrees-awarded-in-library-science-correlates-with-google-searches-for-how-to-hide-a-body](https://tylervigen.com/spurious/correlation/1840_bachelors-degrees-awarded-in-library-science-correlates-with-google-searches-for-how-to-hide-a-body)  
And it's kind of hilarious that now Tyler has made an AI try to make up a "reason" for why these spurious correlations exist. The layers of spurious BS are amazing!

## Recap

So to briefly sum up, here's what we've outlined so far:

1. Data is valuable, because it consists of *values* that contain some kind of *meaning* that relates to something we might care about.
2. But data doesn't tell you anything on its own, for that you need *data analysis*.
3. Data analysis can be broadly broken into two (deeply intertwined) phases: *exploratory* analysis and *inferential* analysis.
4. Exploratory analysis can help you identify patterns, but there is always a level of uncertainty in those patterns, or what we can conclude from those patterns.
5. Statistical methods help us to perform inferential analysis that takes uncertainty into account.
6. Statistical methods are a *complement* to careful, reasoned thinking about a problem, not a *substitute*.

Whew! That's a bunch of steps. Get used to it – mathematical and statistical reasoning often require you to work through several interconnected steps to go from a set of starting assumptions to a final analysis or set of conclusions. Keeping track of the logic is important, because losing our way at any of those steps is a sure way to get lost in invalid and unhelpful analytical dead-ends, or worse, overconfidence in spurious results, like a potential connection between librarians and missing bodies.

## The Shape of Data

Now that we've talked about what data is and what we want to do with it, we need to talk more specifically about the shapes it comes in, because in order to be able to work with data and manipulate it, we need some concepts and terms to help guide us.

## Variables

The most basic unit of data that we will deal with is the *variable*. A variable is a single *dimension* of our data, representing just one aspect or type of measurement. So if we have a bunch of medical records in a data set, one variable may be the weights of patients, another might be resting heart rates, another might be systolic blood pressure, and so on. These are called variables because they *vary* in our data, in contrast with *constants*, which don't vary. We typically don't even think of constants as "data", precisely because they are 100% predictable (or otherwise determined), but we will see how the concept of a constant is an important part of some statistical models.

This is another case where we will have to deal with some ambiguity, or rather, homophony. When we talk about variables in a statistical sense, we mean something else.

But *variables* are where the action is, so to speak. They are the values we take pains to collect, they are the things we are trying to predict. The key is that a given variable is information along a certain *scale*, a single dimension. A collection of data that includes different kinds of information, like height, weight, and blood sugar levels, is not a single variable, but multiple.

Note that just having a variable doesn't tell you how much data you have. A variable with one value is still a variable.

## Types of variables

One of the important things to know about a variable when we start to analyze is what type of information it contains. Depending on what type of data it represents, some methods for analysis may or may not be appropriate or misleading.

We can first categorize data roughly into *numeric* vs. *non-numeric* data. Sometimes the line can blur between these, but this is usually the first consideration. Then, if we consider just numeric data first, that also breaks down into some different types:



- Continuous: like the real number line
- Interval: like the set of integers
- Ordinal: like a ranking
- Nominal: standing in for unordered categories

Let's talk about each of these in turn.

## Continuous data

Continuous numerical data refers to the idea that within the ranges of data, essentially *any* numeric value is possible, up to the number of digits of measurement precision. One example might be weight, where given a sensitive enough scale, you could get potentially any value along the number line.

In practice, virtually any data that is represented down to decimal levels is treated as continuous without any problem. However, people often treat data that is maybe less obviously continuous as still “continuous enough.” This is often fine, but sometimes not, and it's crucial to always keep in mind exactly what kind of data you have, as you consider ways to analyze it.

## Interval data

One type of data that falls in the category of “almost continuous, but not really” is called *interval* data. It's called this because the idea is that the distance between values is meaningful and constant at different parts of the scale. To be clear, continuous data is **also** interval data. The term “interval data” just means the data may or may not be continuous, but can still be treated as even intervals.

One example would be the number of children in a home. You can only have an integer number of children in a given home, and it's an interval quantity, because the difference between 1 and 2 children is the same as the difference between 4 and 5 children: namely 1 child. But as the old joke goes, when you do things like take an average over all households in a population and you come up with something like “the average household has 1.4 children”, it becomes clear that the number

of children is definitely not a continuous measure in reality, though it might be interval data.

Now of course, if you have experience with children in households, you might balk at the idea

## Ordinal data

Loosening things up even more is when we treat numeric data as merely *ordinal*. These are numbers that rank different values, but there is no expectation that it has the *interval* property. In other words, differences between different values may not be equivalent.

One example is rankings. Imagine taking three people from a university tennis team and ranking them 1, 2, and 3. Maybe they are not all that different, but maybe the 1st and 2nd ranked players are closer to each other than the 3rd. Then to drive the point home, imagine adding Rafael Nadal to the mix. Now the distance between the 1 and 2 rankings is hugely different than the distance between the 2 and 3 or 3 and 4 rankings (unless this is an enormously talented college team).

Another very common example in data analysis are ratings such as so-called “Likert” ratings from surveys.<sup>6</sup> These sometime have words associated with them, like “strongly agree” through “strongly disagree”, or they may just be of the kind, “rate this on a scale from 1 to 5”. Because these are judgments, and people have all sort of preconceived or other notions about what these ratings mean, they are very unlikely to be *interval* data. For example, it may be erroneous to assume the difference between a 5-star and a 4-star review is the same as the difference between a 4-star and a 3-star review. Clearly 5 is better than 4 and 4 is better than 3, but *how much better* may not be constant. This is the hallmark of *ordinal* data.

<sup>6</sup> The same typically holds true for other kinds of rating systems, like “star” rating systems like movie reviews, game review, Uber reviews, Yelp reviews, etc.

## Nominal data

Finally, we have the type of data where we think there are differences between the values, but there’s no inherent ranking. If these are numeric data, it’s usually because the numbers just

stand in for category labels, and it's arbitrary which numbers go with which categories.

One example might be eye color in a data base of physical features of people. There is no obvious sense that "brown eyes" is a higher or lower value than "blue eyes," they're just different.<sup>7</sup> To further illustrate, imagine a data base that recorded brown eyes as 1, green eyes as 2, and blue eyes as 3. Then imagine that the data consisted of 20 brown-eyed people, 5 green-eyed people, and 20 blue-eyed people. If you treated this as interval data or (heavens forbid) continuous data, then it might be reasonable to calculate the average eye color, and you would end up with an average of 2, meaning green-eyed people, even though that group was the least numerous. Hopefully you can see how this would be problematic.

Nominal data is also often called "categorical" data.

## Non-numeric data

Non-numeric data really doesn't introduce any new categories, but rather just represents a subset of the above categories. Often non-numeric data is nominal (aka categorical), but sometimes it can be ordinal, like the Likert response categories such as "strongly agree", "somewhat agree", "somewhat disagree", "somewhat agree".

So ultimately, the choice between numeric and non-numeric is a matter of how the data is coded or represented. We can always choose to take non-numeric data and use numbers to stand in for the different (non-numeric) values. What's more important is that we keep in mind that we don't treat all numeric data as continuous or even interval; it's always important to think about what type of data your variable represents before continuing your analysis.

Now, let's talk about what happens when you have multiple variables in a data set.

<sup>7</sup> Can you think of any cases that might be a counter-example? What kind of data or analysis would you need to be doing in order to treat, say, brown, green, and blue eyes as ordinal? What would be determining the order?

## Data Frames

In truth, while you may be able to do some useful things with a single variable, most of the really interesting stuff happens when you have more than one variable. But what shape does that take?

One of the most useful and ubiquitous shapes that data comes in is something called the “data frame.” There are other names for this, like “tables” in a data base context, or a simple spreadsheet in a program like Excel, but the idea is simple: data is in a rectangular format, with rows and columns, such that every row has a value (or a “missing” value) for every column, and every column has a value (or “missing”) for every row.

Furthermore, by convention the columns are *variables*, and the rows are *records* or *observations*. For example, consider a medical data base where we had three variables: height, weight, and (average) resting heart rate. If we had some way to link these, for example we had each of these values for a number of patients, then we have a data frame: a column for each variable, plus a column that identifies a patient, and every row is the data for a single patient.

What if we added time to this, where we had multiple measurements over time across multiple patients? This is where things start to get interesting and choices have to be made. But one way to represent this data might be a data frame that includes an additional column, that of time. Now patients would be represented on multiple rows, so a row is no longer a “patient”, but rather something like “a patient at a specific time”.

But the real answer is that while there are some basic conventions and principles, like representing variables in columns and observations/records in rows, there are often lots of different shapes that one could arrange a given data set into, depending on your purposes. We will see examples of that as we continue in the course, and it is why the ability to re-format and re-shape your data is a critical skill to develop as a data analyst.

For now, we will focus on data frames and use their implementation in R to capture our data, and we will hold on to the

idea that columns roughly correspond to variables, and rows roughly correspond to a set of linked observations.

## First steps in Exploratory Data Analysis

With these preliminary goals, concepts, and definitions in mind, we can start to think about how to approach exploratory data analysis.

### Finding good questions

The person who coined the term “exploratory data analysis” was John Tukey, a famous 20th-century statistician. Tukey was extremely influential in many ways, including bringing back the use of graphics to help with statistical analysis, especially in exploratory analysis. He is also widely credited with coining the term “bit” for the basic unit of computing memory/storage, and he has the first published use of the word “software.”

But for our purposes in this course, Tukey leaves behind a series of observations and ideas that help guide us through the exploratory data analysis process.

First, a little humility is in order:

The combination of some data and an aching desire  
for an answer does not ensure that a reasonable  
answer can be extracted from a given body of data.  
– John Tukey

To paraphrase: just because you *really want* to find an answer to a question and you have some data – even a LOT – of data, there may not be a reasonable answer to find.

This is good, sober advice to help us keep from going to astray just because we are committed to finding something in a data set. But it also subtly emphasizes the importance of questions. Another quote that nicely complements this one:

An approximate answer to the right problem is worth a good deal more than an exact answer to an approximate problem. – John Tukey

Paraphrase: getting the right questions and exploring the right problems is more important than getting “perfect” answers to unimportant or misunderstood questions/problems.

Here’s an example in terms of modern app development that illustrates both of these ideas. Imagine a hypothetical app called SayWhat that sells itself as a language-learning app. The developers of that app are constantly adjusting and working on it. But how do they decide what is a “good” change to the app? They surely have a huge amount of data to work with. So shouldn’t they be able to use that data to assess whether one version of the app is “better” than another?

Well, it comes down to how you define the problem: what does “better” mean? The version of “better” that might occur to most users of the app would be an app that helps people to more quickly reach functional levels of language proficiency. But here’s the problem, SayWhat doesn’t collect any good measures of real-world functional language proficiency. What they have is people’s performance on the app itself, and the app itself is designed to keep people engaged, not to turn them off by “failure.” So “better” from the developers’ perspective is more like, “the version that keeps users in the app longer,” not “the version that helps users learn faster to the point where they don’t need the app.” But more to Tukey’s point, the kind of data collected by SayWhat – even though there’s a lot of it! – is unlikely to give even an approximate answer to the right question.

Whether this hypothetical scenario reflects real-world language-learning apps is left as an exercise for the reader.

Back to the topic at hand, where I’m going with all this is that good exploratory data analysis helps guide you to good questions.

## Expectations and surprise

The way that exploratory analysis helps inform your questions comes from a back-and-forth and contrast between checking expectations and finding surprises. What do I mean by that?

When you start to examine a data set, it's helpful to start with thinking explicitly about your expectations of what you might find (and what you *should* find), while still being open to being surprised.

The first Code Practice and Challenge ask you to do this by thinking about what kinds of values you might find in your data, before you actually check and calculate them. For example, if you had data on Yelp reviews, you'd expect to see values from 1 to 5, because that's how their rating works. If you saw values that went up to 100, you might have more questions about what the data actually represents. Similarly, if you had a data set that was supposed to have weight data of adults, but you saw values below 30 lbs. or above 800 lbs., you might have questions about the validity of some of that data.

This is an example of a process of checking basic assumptions.

But beyond this, we might have other assumptions that we didn't necessarily even think about. A good exploratory analysis will help you confront those assumptions, which may turn out to be an important step towards a better analysis. There may be a pattern you wouldn't have considered before you checked. There may be a set of outlier data that you hadn't expected that could open up new questions. And so on.

In short, a good exploratory analysis aims to confirm basic assumptions, but also to help find surprises that might help you revise your questions about what you can even answer in your data.

## Numbers and graphics

One of Tukey's most convincing claims is the essential nature of graphical methods – data visualization – to exploratory data analysis. Getting numerical and statistical summaries is also

helpful, and that’s where we are starting. But beginning in the next unit, we will add graphical techniques as well. We will of course only get the tip of the iceberg in terms of data visualization, but we will cover enough to be useful, particularly towards supporting regression analysis.

But in the present unit, we will simply start with a few basic numerical summaries, as described in the R Basics documents, and in the first Code Practice and Challenge.

## Basic numerical summaries

When you start to explore data, you should consider – at a *minimum* – several aspects of each numerical variable:

- What at the *extreme* values?
- What are the *central* values?

*Extreme* values are simply the smallest (minimum) and largest (maximum) values for a particular variable. This is collectively called the *range*. What values does your data cover? Knowing this is an important first step towards confirming and/or challenging your expectations about your data.

*Central* values have a bit more complexity behind them, which we will get into more deeply in the next unit. For now, we’ll consider two central values, the *mean* and the *median* of a variable.

The *median* is the “middle” value. That is, if you take all the values you have, sort them, and line them up from small to large, the median value is right in the middle, where roughly half the values are smaller and half are larger.<sup>8</sup> This is useful because it helps you see where your data “split”, regardless of how far away the extremes are.

The *mean* is another word for *average*, and is the sum of all the values divided by the number of values. This number is sometimes referred to as the “expected value” of the variable. We will explain this more fully in the next unit, but the idea is that all else equal, if you collected another data point for that variable, the mean is your “best guess” for what that data point is likely to be.

<sup>8</sup> If there are an even number of values, there isn’t exactly a “middle” number, and different methods compute the median in different ways. In R, it’s the mid-point between the two middle numbers, which is the same as the average of the two middle numbers.



How far the mean and median are from each other tells you something about the shape of the overall distribution of the variable, but again, we will explore this more soon. In the meantime, just keep in mind that if the mean and median are very different, it warrants some follow-up in the exploratory analysis.

## **Next steps**

From this document, you should continue on to the R Basics notebooks. Come back to this document when you are working on your Knowledge Check, and when you need to review concepts.